

REU-HPC: Week-1 Initial Meeting

Welcome to the REU program! This document summarizes our initial discussion, project expectations, and recommended resources to help you get started.

Research Project

Our research focuses on performance monitoring, profiling, and performance analysis tools for High Performance Computing (HPC) systems.

The primary research tools you will be working with are:

- **PEAK (Performance Evaluation and Analysis Kit)**
- **FLAME (Function-Level Adaptive Monitoring Engine)**

These tools are being actively developed as part of a broader effort to build cost-adaptive performance monitoring solutions for large-scale HPC applications.

Working Hours and Location

- Regular working schedule: Monday–Friday (flexible working hours)
- REU participants are expected to collaborate **on-site at TACC every Wednesday**
- Additional workspace is available in the **POB VisLab (Room 2.404)** throughout the week

Communication

- **Email**
Albert: alu@tacc.utexas.edu
Ian: yinzhi.wang.cug@gmail.com
- **Slack**
REU-HPC Slack Channel

Please use email and Slack for questions, progress updates, and scheduling discussions.

Shared Documents (google drive)

https://drive.google.com/drive/folders/1wpJQ217PJ0si1z-YLDTm9E7k9Gulyiho?usp=drive_link

Group Meetings

Preferred meeting day and time?

Weekly Progress Report

Beginning next week, please send a brief **half page progress report** before each group meeting. The report does not need to be formal. Bullet points are perfectly acceptable.

Suggested contents:

What I worked on this week

Tasks completed, software installed, documentation reviewed, experiments performed, ...

What I learned

New HPC concepts, tools or techniques learned, technical challenges encountered, ...

Questions / Issues

Problems that need discussion, topics requiring clarification

The purpose is to help track progress and identify areas where additional guidance may be helpful.

Week 2 Objectives (June 8 – June 12)

The primary goal for week 2 is to become familiar with:

- TACC systems and basic HPC workflows
- Linux command-line usage
- PEAK as the primary research tool

At this stage, focus on understanding the environment rather than producing research results.

HPC Prerequisites

Please become familiar with the following topics:

TACC Accounts and Access

- TACC account setup
- Project and allocation access
- SSH login
- Multi-factor authentication (MFA)

Linux and HPC Basics

- Linux command line
- File management (create, copy, edit, delete, transfer, etc.)
- Environment modules (environment variables and package modules)
- Batch job submission
- Job status monitoring

Recommended Learning Resources

- ***Intro to Linux and HPC***
<https://introtolinuxhpc.readthedocs.io/en/latest/>
- ***TACC Training Videos***
https://www.youtube.com/@TACC_training/videos
- ***TACC User Resource Portal***
<https://utrc.tacc.utexas.edu/resources/>

TACC Systems Documentation

- ***Frontera***
<https://docs.tacc.utexas.edu/hpc/frontera/>
- ***Stampede3***
<https://docs.tacc.utexas.edu/hpc/stampede3/>
- ***Lonestar6***
<https://docs.tacc.utexas.edu/hpc/lonestar6/>
- ***Vista***
<https://docs.tacc.utexas.edu/hpc/vista/>

PEAK Reading Assignment

Before our next group meeting, please spend some time exploring PEAK independently.

Start with:

- **PEAK GitHub Repository**
<https://github.com/peak-team/peak>
- **PEAK Documentation**
<https://deepwiki.com/peak-team/peak>
- **Research Papers**
PEAK: a Light-Weight Profiler for HPC Systems
<https://dl.acm.org/doi/10.1145/3624062.3624143>

PEAK: Cost-Adaptive Profiling in a Heartbeat
<https://dl.acm.org/doi/10.1145/3731599.3767521>

Next Meeting Discussion

During the next group meeting, give a short informal presentation (~5 minutes) covering:

- Your understanding of PEAK
- The overall purpose of the tool
- Major components and architecture
- Features they found interesting
- Questions or observations from reading the documentation and source code

The goal is not to fully understand every detail, but to begin building familiarity with the project and identify areas for further exploration.

We look forward to working with you throughout the summer and helping you develop both HPC skills and research experience.